

Figure 1: HUMAN-gate state transition. Only a valid HUMAN proposal followed by an explicit approval reaches the simulated ledger; every invalid, rejected, or expired path terminates without actuation. The ledger and actuation are simulated, not a production O-RAN deployment.

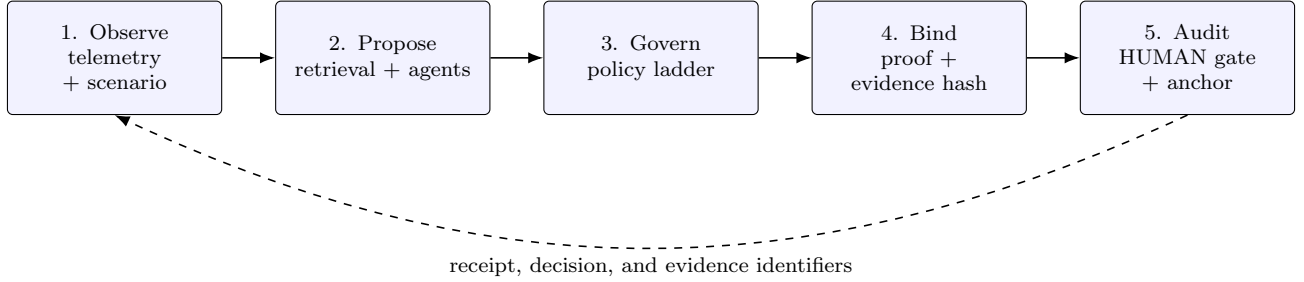


Figure 2: Five-stage journal schematic. Model output remains an untrusted proposal until the external policy ladder admits it; the full 35-node n8n canvas and node-level configuration belong in the supplement.

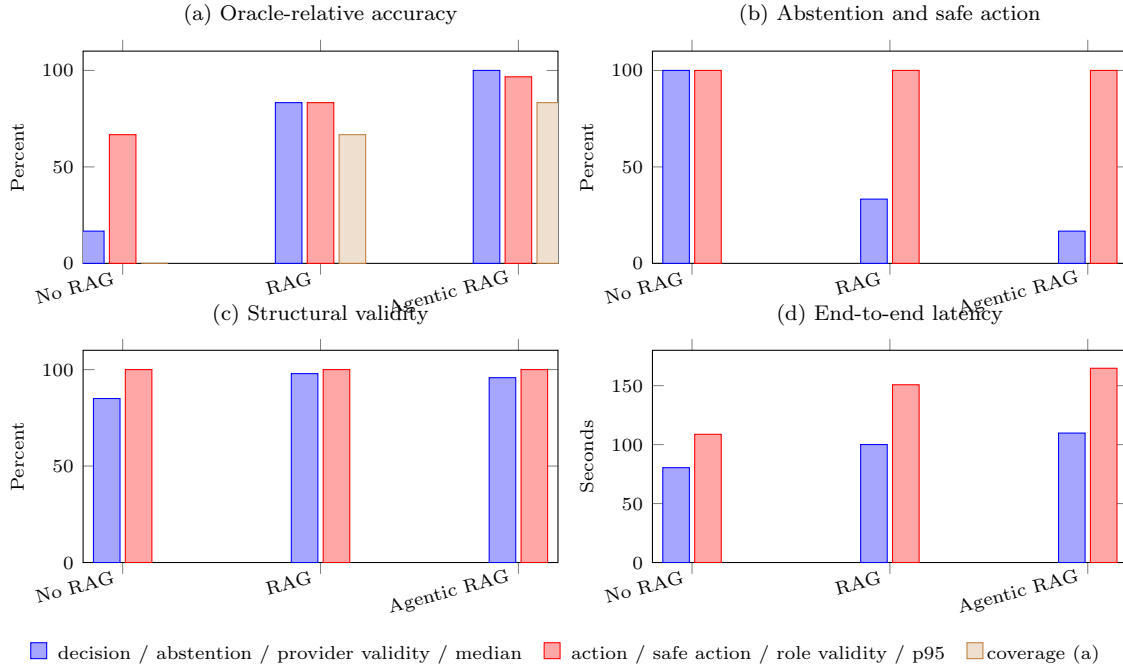


Figure 3: Consolidated frozen-oracle results (60 runs per mode). In (b), the series are abstention and safe-action rate; in (c), provider-schema and role-schema validity; in (d), median and p95 latency. Accuracy is relative to the synthetic oracle, and latency targets Non-RT rather than Near-RT control.

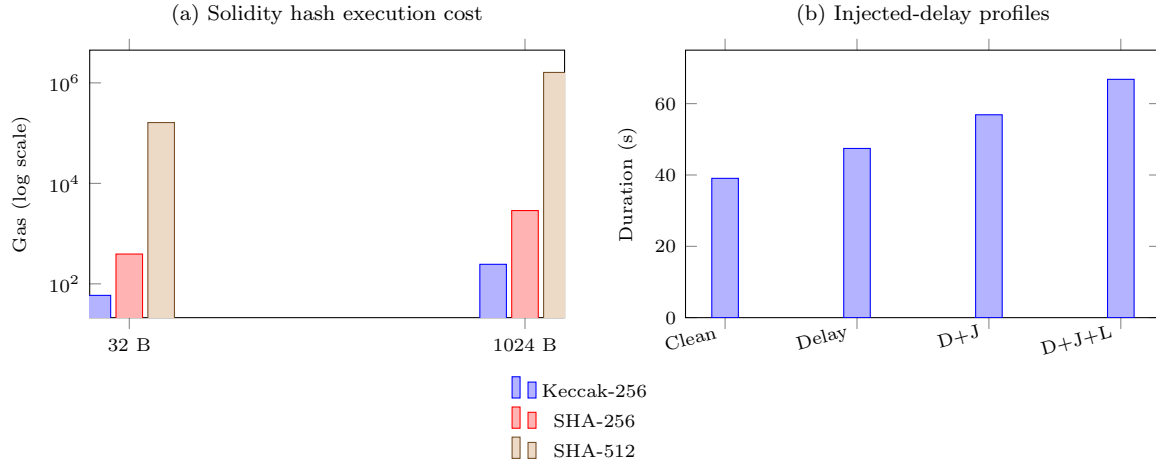


Figure 4: Consolidated cost and stress results. Panel (a) uses a logarithmic axis because the Solidity SHA-512 implementation is orders of magnitude more expensive; panel (b) reports observed workflow duration under cumulative delay (D), jitter (J), and loss/retry (L) injection. These experiments have distinct workloads and should not be interpreted as a single end-to-end deployment benchmark.